

7. LRU Page Replacement Algorithm

FACE Prep

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LRU Page Replacement Algorithm

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Sample Input 1

```
5
1 2 3 4 5
5
```

Sample Output 1

Enter number of pages in reference string:

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, m;
5     printf("Enter number of pages in reference string: \n");
6     scanf("%d", &n);
7
8     int ref[n];
9     printf("Enter the reference string: \n");
10    for (int i = 0; i < n; i++) {
11        scanf("%d", &ref[i]);
12    }
13
14    printf("Enter number of frames: \n");
15    scanf("%d", &m);
16
17    int frames[m];
18    int lastUsed[m];
19    for (int i = 0; i < m; i++) {
20        frames[i] = -1;
21        lastUsed[i] = -1;
22    }
23
```

Test Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2

Run Sample Cases

Run All Cases

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The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Sample Input 1

```
5
1 2 3 4 5
5
```

Sample Output 1

Enter number of pages in reference string:

Select Language: C (GCC 9.2.0)

```
36 }
37
38 if (found != -1) {
39     hits++;
40     lastUsed[found] = i;
41 } else {
42     faults++;
43     int replaceIndex = -1;
44
45     for (int j = 0; j < m; j++) {
46         if (frames[j] == -1) {
47             replaceIndex = j;
48             break;
49         }
50     }
51
52     if (replaceIndex == -1) {
53         int lruIndex = 0;
54         int minTime = lastUsed[0];
55         for (int j = 1; j < m; j++) {
56             if (lastUsed[j] < minTime) {
57                 minTime = lastUsed[j];
58                 lruIndex = j;
59             }
60         }
61         replaceIndex = lruIndex;
62     }
63     frames[replaceIndex] = ref[i];
64     lastUsed[replaceIndex] = i;
65 }
66 }
```

Test Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2

Run Sample Cases

Run All Cases

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Input Format

1. The first line of input contains an integer representing the number of pages in the reference string, n .
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Sample Input 1

5

1 2 3 4 5

5

Sample Output 1

Enter number of pages in reference string:

Select Language : C (GCC 9.2.0)

```
52 if (replaceIndex == -1) {
53     int lruIndex = 0;
54     int minTime = lastUsed[0];
55     for (int j = 1; j < m; j++) {
56         if (lastUsed[j] < minTime) {
57             minTime = lastUsed[j];
58             lruIndex = j;
59         }
60     }
61     replaceIndex = lruIndex;
62 }
63 frames[replaceIndex] = page;
64 lastUsed[replaceIndex] = i;
65 }
66 }
67 }
68 }
69 printf("\nTotal Page Faults: %d\n", faults);
70 printf("\nTotal Page Hits: %d\n", hits);
71 }
72 return 0;
73 }
74 }
```

[Test Cases](#)[Run Sample Cases](#)[Run All Cases](#)

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2